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CCC 821 – Algorithmic Thinking

Assignment

Question 1

A building has 100 floors. One of the floors is the highest floor an egg can be dropped from without breaking. If an egg is dropped from above that floor, it will break. If it is dropped from that floor or below, it will be completely undamaged and you can drop the egg again. Given two eggs, find the highest floor an egg can be dropped from without breaking, with as few drops as possible.

Answer:

We see, in this question that if the egg breaks from a floor, say, *n*, then it will also break from any floor that is higher than *n*.

Looking at this question, I was reminded of the Binary Search algorithm that we had discussed in class. If there was no constraint on the number of eggs we could use, we would have gone ahead and dropped the first egg from the 50th floor. Had it broken, our area of measurement would have been reduced to floors 1-49, to check the highest floor from which the egg could be dropped, without breaking.

In the case that it had NOT broken when we dropped from the 50th floor, we would have been left with floors 51-100 to check out. In this way, we could have carried on testing by dropping an egg from the middle floor from amongst the range of floors left in consideration. This would result in halving the floors to be considered with each drop and hence, it is evident that we would require $\boldsymbol{\log_2 100}$ eggs, which means that we'd require **7** eggs. Unfortunately, we do not have 7, and Rajeev sir has allowed us to play around with only 2. It would not be much of a challenge otherwise.

To get to what we'd do with two eggs, here is what I would do if I had just one egg. Obviously, I'd start with the safest, the lowermost, floor number 1, and keep moving upwards floor by floor, until the egg would finally break at, say, floor *n*. Then, it would be safe for us to say that the $(n-1)^{th}$ floor is the one we're looking for. In this scenario, the worst case would be that the egg does not break on any of the floors and that would require **100** drops from our side to test.

But now, we have two eggs. What we could do, is drop one egg from the 50th floor, and since we always consider the worst case while analysing any algorithm, we assume that it breaks and we're left with just one egg. Now, we have floors 1 through 49 left to test. So the maximum number of drops required from our side is **50** in this scenario, which is an improvement from the One Egg case, but is still a poor algorithm.

The new idea which struck me after I came up with this pathetic excuse for a solution was to drop the first egg at intervals of 10 floors – 10, 20, 30… and so on. Let us say that the egg did not break until floor 50, but broke when dropped from the 60th floor. This gives us an idea that the solution lies somewhere between the 50th and the 60th floor. So we go back to the one egg strategy and drop the egg from the 51st floor and move upwards floor by floor, and we'd get the solution easily. The worst case for this strategy would be when the solution is the 99th floor. That would require drops from our side from the 10th, 20th, 30th… 100th floor = 10 drops and 9 additional drops from the 91st, 92nd, 93rd … 99th floor.

So the worst case in this solution is **19 drops**, which seems like a pretty good solution considering that we've just improved from 50 drops.

DROP NUMBER	FLOOR
1	10th
2	20th
3	30th
4	40th
5	50th
6	60th
7	70th
8	80th
9	90th
10	100th

The only problem is that we notice that when the solution is on a lower floor, then the maximum number of drops required to determine is lesser and keeps increasing as the solution becomes higher. For example, if the solution is between the 10th and the 20th floor, then the worst case is just 11 drops, whereas we saw that if it is between the 90th and the 100th floor, then the worst case is 19 drops.

So, I tried looking for a way in which I could sort of equalise the number of drops required for all case, across the board, and makes things more uniform. I noticed that if we drop the first egg at the nth floor, then we have to sweep the previous *(n-1)* floors to find the solution, if the egg breaks at the nth floor. However, if it does not break, then we should ideally jump (*n-1*) floors and not *n* floors as we would be using an extra drop in this jump and we should try and compensate for it in the second half of the search, in order to not allow our worst case for this scenario to be worse than the previous one. So, the second drop would be from the floor given by the formula *n+(n-1)* and similarly, the drop after that would be from the $[n+(n-1)+(n-2)]^{th}$ floor.

Since the total number of floors is **100**,

We see,

$(n) + (n-1) + (n-2) + (n-3) + \ldots + 2 + 1 >= 100$

Therefore, $(n)(n+1)/2 >= 100$

- ⇨ $n^2 + n - 200 >= 0$
- ⇨ $n >= 13.65$
- ⇨ $\underline{\mathbf{n = 14}}$

So, we now have the optimal solution to the given problem. The first drop would be from the 14th floor, the second from the 27th, and so on, as shown in the table.

DROP NUMBER	FLOOR
1	14th
2	27th
3	39th
4	50th
5	60th
6	69th
7	77th
8	84th
9	90th
10	95th
11	99th
12	100th

Now, while this is a perfectly good solution that requires a maximum of 14 drops in all possible cases, I just noticed that if the solution is 100, we can get to it in just 12 drops, whereas we require 14 for all other cases. So if we just tweak the 11th drop to be from the 98th floor instead if the 99th, we can make the number of drops required for floors 97-100 to become 13 as opposed to the earlier case, which was 13 for 97, 14 for 98-99 and 12 for 100. This is just my personal opinion to shift the drop and does not really make much of a difference to the solution, but just serves to make it more uniform.

We see below the final answer and the sequence of drops to be followed to determine the highest floor from which the egg can be dropped, without breaking.

DROP NUMBER	FLOOR
1	14th
2	27th
3	39th
4	50th
5	60th
6	69th
7	77th
8	84th
9	90th
10	95th
11	98th
12	100th

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Question 2

How many Cricket balls can fit in a school bus? Explain your answer with suitable assumptions.

Answer:

Please Note:

I am not using any standard dimensions in this question, the measurements that I take in this question are simply based on my observations of the dimensions of a school bus and that of a cricket ball, and hence, may not be accurate.

Assumed dimensions of a school bus:

- ⇨ Length = 40 feet
- ⇨ Width = 10 feet
- ⇨ Height = 10 feet

(I have taken all these estimations using my own height as a yardstick in my head and comparing the dimensions relative to it. I have made the estimations in feet because I am more comfortable thinking in terms of the same.)

Now, for the sake of ease of calculation and clarity, we should convert the dimensions to metres, and round them off for convenience.
So,

- ⇨ Length = 40 feet = Approx. 12 m
- ⇨ Width = 10 feet = Approx. 3 m
- ⇨ Height = 10 feet = Approx. 3m

Now, we must take into account that the internal height of the bus is obviously lesser than 3 m, taking into account the ground clearance of the bus. And since most of the school buses we see are not those low-floor ones, my guess is that the internal height is **2.5m**.

So,

- ⇨ Total internal volume = L x B x H
  = 12 x 3 x 2.5
  = **90 cubic metres**

However, it would be incorrect to assume that the entire internal volume of the bus is empty as it is furnished with seats, luggage racks, the driving paraphernalia, and other such stuff which you'd expect to see on the inside of an empty school bus.

Since there is no way of knowing the exact volume occupied by it, a reasonable estimate, according to me, would be that **80-85%** of the calculated internal volume is indeed empty.

So effective internal volume works out to be approximately **75 cubic metres.**

Coming to the cricket ball,

- ⇨ Estimated ball diameter = 8 cm
  = 0.08 m

Now, to make calculation simpler, I am going to assume that the volume occupied by a single ball is equal to the volume of a cube with side equal to the diameter of the ball. This helps us account for the empty spaces (voids) that would definitely be created when the balls are stacked together.

Therefore, volume occupied by one ball = $(0.08)^3$

= 5.12×10^{-4} cubic metres

= Approx. **5×10^{-4} cubic metres**

Hence, we can calculate the number of cricket balls that can fit in a school bus by simply dividing the effective internal volume of the bus by the effective volume of a single cricket ball.

- ⇨ Number of cricket balls that can be fit = $90/(5\times10^{-4})$
  <u>= 1,80,000 balls</u>

*Therefore, according to my estimations, **<u>One Lac and Eighty thousand cricket balls</u>** can be fit into a school bus, which seems like an awful lot, really. But hey, it's numbers, and the math doesn't lie.*

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Consider the following example in which marks of 6 students are given:

50, 52, 54, 56, 60, 94

In this case,

- ⇨ *Mean = 81*
- ⇨ *Median = 75*

Therefore, we see that the mean is actually greater than almost all of the values whereas the median is a good representation of the central tendency.

This rules out **MEAN** from our consideration, because in a case like land prices in Delhi, there are bound to be outliers, such as the price of land in posh localities or in highly sought after commercial belts. They would distort the mean and render it as an ineffective tool for measuring the central tendency.

Now, let us see why we cannot use **MODE**. There are three simple cases which may occur, according to me, because of which we should not use it:

1) *In a continuous distribution of data such as this (price of land), it is possible that multiple entries of the same price do not come up (mode does not exist), or may be very few in number.*
   This renders the mode pretty much useless in this scenario.
2) *There may exist two (or more) modes in the data set, i.e., multiple prices may have the same highest frequency. In such a case, it is unclear how to proceed and would lead to greater confusion than actual utility.*
3) *The mode may be present near the lower or upper extreme of the data set of prices, and in that case, it would not give us an accurate representation of the central tendency.*

- ⇨ Having eliminated both mean and mode satisfactorily, it is now safe for us to say that the **MEDIAN** would give us the best representation of central tendency, since the median, by its very definition, gives us the middle value in a data set, and in a continuous distribution of data, such as the one we have right here, proves to be an extremely valuable and accurate tool to measure the central tendency.

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We see,

The number of flowers in the field doubles every day, so if we were to go the previous day, we'd find that the number of flowers keeps on getting halved. Therefore, the field would be half full **at the end of 44 days.**

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